THESIS EXCERPT

Personalized Insulin Dosing Support Through Structured Data Visualization and Pattern Analysis: A Foundation for ML and RL-Assisted Pharmacotherapy in Type 1 Diabetes

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Abstract

Type 1 Diabetes (T1D) represents one of the most demanding chronic conditions in modern medicine, requiring patients to make continuous, high-stakes decisions about insulin dosing under conditions of significant physiological variability. Existing management tools rely primarily on simplified carbohydrate-counting rules that fail to capture the full complexity of glucose-insulin dynamics — including the effects of macronutrient composition, meal timing, physical activity, circadian insulin sensitivity variation, and compounding pharmacodynamic effects over time.

This thesis presents DiaNavigator (subsequently evolved into DiaTwin), a modular clinical decision-support platform that integrates a physiologically grounded pharmacodynamic engine, real-time glucose estimation, continuous glucose monitor (CGM) integration, and a multi-role clinical interface into a unified digital platform. The core physiological model employs gamma-distribution curves — parameterised from the German S3 Clinical Guideline (AWMF-Registry 057-013) — to model both carbohydrate absorption and insulin activity with biophysical accuracy across 12 insulin types and 5 absorption profiles.

The platform implements a 9-factor personalised insulin dosing algorithm incorporating insulin-on-board (IOB), meal-on-board (MOB), protein and fat delayed-absorption corrections, time-of-day insulin sensitivity factors, activity coefficients, and disease-specific multipliers. A cumulative baseline tracking system models the T1D physiological reality that glucose effects stack and persist rather than self-correct. Technical evaluation demonstrated system stability, sub-second calculation latency, and scalability under concurrent load. Clinical expert review confirmed the appropriateness of the recommendation logic and explainability framework.

The data architecture is explicitly designed as a foundation for supervised and reinforcement learning integration, providing structured, high-quality datasets suitable for adaptive insulin dosing policy learning. This thesis establishes the theoretical and technical groundwork for a full digital twin of T1D physiology, with identified extensions including hormonal and menstrual cycle modelling, stress and sleep state detection, and RL-based personalised AID optimisation.

Keywords: *Type 1 Diabetes, Digital Twin, Insulin Pharmacodynamics, Gamma Distribution, Machine Learning, Rei*

Chapter 1: Introduction

1.1 Background

Diabetes mellitus is a chronic metabolic illness that has been known to humanity for millennia, yet remains one of the most serious public health challenges of modern times. The global prevalence of diabetes continues to increase at an alarming rate, with projections indicating further growth over the coming decades. Among the various types of diabetes, Type 1 Diabetes (T1D) occupies a special position: it is an autoimmune condition involving the destruction of pancreatic beta cells, resulting in absolute absence of endogenous insulin production. Unlike Type 2 diabetes, T1D typically presents with rapidly occurring symptoms and is frequently diagnosed in childhood or adolescence, demanding lifelong intensive management.

Living with T1D involves far more than daily injections or pump therapy. Patients must continuously monitor blood glucose levels, adjust insulin regimens in response to meals and physical activity, and remain vigilant against acute complications such as hypoglycaemia and diabetic ketoacidosis. The complexity of T1D management is further elevated by the fact that insulin requirements can change dramatically depending on stress, illness, hormonal changes, and the absorption characteristics of specific foods. This variability creates persistent uncertainty and psychological burden for patients and their families. Even with advances in insulin formulations, delivery devices, and glucose monitoring technology, a significant proportion of people with T1D fail to achieve or maintain optimal glycaemic control.

The economic cost of diabetes management is substantial. T1D treatment involves both direct costs — insulin, monitoring devices, and medical consultations — and indirect costs including lost productivity, disability, and premature mortality. Advanced technologies such as CGM systems and automated insulin delivery (AID) pumps have improved outcomes for many, but access inequalities persist across socioeconomic and geographic contexts. The psychological toll is equally significant, with a large proportion of T1D patients reporting diabetes distress, anxiety, and depression attributable to the exhausting demands of self-management.

1.2 Motivation

The motivation behind this research arises from the gap between the theoretical capabilities of advanced diabetes technology and the practical realities of daily T1D management. Although CGMs and insulin pumps have achieved significant penetration in high-resource contexts, their usage is far from universal, and they introduce new complexities: device calibration, data overload, and the need for ongoing technical support. In practice, many patients still rely on manual data entry, memory, and intuition to make critical insulin dosing decisions.

Consider the daily experience of a young adult with T1D: an active university student with access to a CGM and modern insulin pump who nonetheless struggles to predict and respond to the interacting effects of food composition, exercise, stress, and hormonal variation on her blood glucose. Template algorithms and fixed dosing protocols frequently fail to capture this complexity. The invisible cognitive and emotional labour of continuous self-management — the fear of complications, the mental overhead of every meal and activity decision — can become overwhelming. This experience is not exceptional: it is shared by millions of T1D patients worldwide who encounter formidable obstacles to optimal care despite recent technological

advances.

These real-world realities motivate the development of a new generation of diabetes management tools: platforms that are not merely technologically sophisticated, but that adapt to the actual dynamics of daily life, scale across different resource settings, and empower both patients and clinicians. This research is driven by the goal of building a fully digital ecosystem that combines detailed, individualised data capture with transparent, clinically applicable guidance — within a modular architecture capable of evolving alongside advances in artificial intelligence.

1.3 Research Question and Objectives

The central research question guiding this thesis is:

> *How can a comprehensive digital platform, integrating personalised algorithms, advanced data visualisation, and machine learning readiness, improve insulin dosing accuracy and diabetes management outcomes for individuals with Type 1 diabetes, while maintaining clinical safety and physician oversight?*

The study defines five interdependent objectives:

– Develop a modular, expandable digital platform enabling patients to record detailed data on meals, insulin doses, physical activity, and blood glucose in an intuitive and low-burden manner.

– Incorporate extensive physiological modelling that transcends traditional carbohydrate counting to capture the effects of macronutrient composition, meal timing, activity, and inter-individual variation in insulin sensitivity.

– Provide safe, explainable insulin dosing recommendations, ensuring that both patients and clinicians can interrogate the reasoning behind every suggestion.

– Architect the data infrastructure to support future integration of machine learning and reinforcement learning algorithms, enabling evolution from rule-based to adaptive data-driven decision support.

– Conduct rigorous technical, clinical, and user experience evaluation to assess system performance, scalability, and real-world feasibility — with particular attention to physician oversight and clinical accountability.

Chapter 3 Excerpt: Physiological Modelling & System Architecture

This excerpt presents the core physiological modelling approaches and system architecture relevant to the digital twin simulation work. The full chapter covers system design philosophy, data management architecture, clinical oversight mechanisms, and future AI integration architecture.

3.7.1 Mathematical Models for Insulin Dynamics

The pharmacodynamic engine underpinning the platform models insulin activity using gamma-distribution curves rather than the simplified linear or exponential approximations employed by most consumer diabetes applications. Parameters — onset, peak, and duration — are derived directly from Table 9 of the German S3 Clinical Guideline for T1D Therapy (AWMF-Registry 057-013, Version 5.0), covering 12 insulin types including ultra-rapid analogues (Fiasp, Lyumjev), rapid-acting analogues (NovoRapid, Humalog, Apidra), short-acting human insulin, intermediate-acting NPH, long-acting basal analogues (Toujeo, Tresiba, Levemir), and mixed formulations.

The gamma distribution captures the physiologically accurate shape of insulin activity curves: the onset lag attributable to absorption from the subcutaneous depot, the peak of activity, and the gradual trailing tail of residual effect. Insulin on Board (IOB) is calculated as the integral of remaining activity across all doses administered within the preceding 24 hours, providing a real-time estimate of active insulin and its projected glucose-lowering effect. This IOB value is subtracted from proposed doses to prevent dangerous insulin stacking — a critical patient safety feature specific to T1D physiology.

3.7.2 Meal Impact and Carbohydrate Absorption Models

Carbohydrate absorption is modelled through five physiologically-grounded profiles, each represented as a gamma distribution curve parameterised to match gastric emptying kinetics and intestinal absorption rates:

Profile	Onset	Peak	Duration	Example Foods
Very Slow	~20 min	~3 h	~8 h	Legumes, fatty meals
Slow	~15 min	~2 h	~6 h	Whole grains, dairy
Medium	~10 min	~1.5 h	~4 h	Standard mixed meals
Fast	~5 min	~1 h	~3 h	White bread, fruit
Very Fast	~2 min	~0.5 h	~2 h	Glucose tablets, juice

Table 1. Carbohydrate absorption profiles and pharmacodynamic parameters.

Beyond carbohydrate content, the model incorporates protein and fat contributions to postprandial glucose — features absent from conventional carbohydrate-counting approaches. Protein is converted to a glucose equivalent via a configurable protein_factor (default 0.5), reflecting gluconeogenic potential. Fat is modelled via a fat_factor (default 0.2) that extends absorption duration and delays the postprandial glucose peak — particularly relevant for high-fat meals such as pizza or full-fat dairy where conventional counting systematically underestimates insulin requirements.

3.7.3 Multi-Factor Personalised Dosing Algorithm

The insulin dose recommendation integrates ten individualised factors:

Factor	Description
Carbohydrate Dose	Total carbs / insulin-to-carb ratio (ICR), patient-specific
Protein Correction	Protein x protein_factor added to carb dose
Fat Correction	Fat x fat_factor (slows absorption, extends insulin need)
Meal Timing Factor	Breakfast +20%, dinner -10% vs. standard (dawn phenomenon)
Time-of-Day Factor	Morning (06-10h): +20%; daytime: x1.0; late night: -10%
Absorption Modifier	Very slow x0.6 to very fast x1.4
Activity Coefficient	Vigorous -20% to sedentary +20% (insulin sensitivity)
Disease Factor	T1D x1.0; T2D x0.8; gestational x1.2; insulin resistance x0.7
IOB Correction	Active insulin subtracted to prevent dose stacking
Correction Dose	(Current BG - Target BG) / correction factor

Table 2. Multi-factor insulin dose calculation components.

3.7.6 & 3.7.7 Reinforcement Learning and Digital Twin Strategy

The platform architecture is explicitly designed as a foundation for reinforcement learning (RL)-based insulin dosing policy optimisation. The physiological model functions as a simulation environment: patient state is represented as a tuple of current glucose, IOB, MOB, activity level, time of day, and personalised constants. An RL agent can interact with this environment, proposing dosing actions and receiving reward signals based on subsequent glycaemic outcomes (time-in-range, hypoglycaemia avoidance, glucose variability).

The digital twin concept extends this further: by continuously calibrating the physiological model parameters against real patient data — CGM readings, logged meals, and insulin doses — the system generates a personalised simulation that reflects individual metabolic behaviour rather than population-average pharmacokinetics. This enables prospective scenario modelling (e.g., predicting the glucose impact of a planned meal before eating) and retrospective analysis (identifying why a glucose excursion occurred). The identified extensions for future development include hormonal and menstrual cycle modelling, sleep state detection from wearable signals, and stress-induced insulin resistance quantification — all areas where current AID systems demonstrate systematic gaps.

Chapter 5: Conclusion

5.1 Summary

This thesis has addressed the longstanding and complex challenges of Type 1 diabetes management by developing a first-of-its-kind data-driven platform — DiaNavigator (subsequently evolved into DiaTwin) — that integrates advanced physiological modelling, transparent visualisation, and long-term machine learning capability within a modular, clinically focused architecture. Beginning with a critical analysis of the current T1D management landscape, the study identified fundamental limitations of established insulin dosing strategies, siloed data collection, and the absence of individualised adaptation in existing systems.

The platform was subsequently developed to enable comprehensive recording of meals, insulin doses, activity, and blood glucose, while providing interpretable and explainable recommendations suitable for real-world clinical scenarios. The system architecture prioritises flexibility and clinical safety, ensuring the platform can evolve alongside advances in artificial intelligence while maintaining physician oversight as an inviolable safeguard. Comprehensive technical and user experience testing confirmed system stability, effectiveness, and clinical applicability — establishing a robust baseline for future innovation.

5.2 Research Findings

The study produced several important findings. Technically, the platform successfully operationalised complex insulin dosing logic — incorporating individualised macronutrient adjustments, meal absorption rates, physical activity effects, and time-based insulin sensitivity variation — within a modular architecture that supports real-time calculation, rapid expansion, and compatibility with standard hardware. Clinically, expert review confirmed the appropriateness of the evidence-based recommendation logic, particularly the CGM calibration and individualised parameter correction mechanisms. Visualisation tools enhanced patient comprehension and clinical decision-making by making the pharmacodynamic effects of meals, insulin, and activity transparent and actionable.

Importantly, the data architecture demonstrated readiness for machine learning integration, with well-structured, high-quality datasets suitable for both supervised and reinforcement learning approaches. This establishes the foundation for future development of adaptive, predictive analytics capable of further personalising insulin dosing and optimising glycaemic outcomes. The system also demonstrated practical feasibility through resource-efficient design and user interfaces that support deployment across high- and low-resource settings.

5.3 Limitations

Several limitations accompany this research. Most significantly, large-scale clinical trials with diverse real patient populations have not yet been conducted to validate long-term health outcome effects. The current evaluation, while extensive, was based on simulated data and expert opinion — valuable but not fully reflective of the dynamics encountered in real clinical practice. The physiological models, while substantially more sophisticated than conventional static approaches, remain simplifications of the full complexity of human metabolism.

Integration with EHR systems, external medical hardware, and broader healthcare infrastructure — while architecturally supported — was not implemented or tested in production environments, limiting assessment of real-world interoperability. The applicability of the system to paediatric populations, geriatric patients, and patients with significant comorbidities requires dedicated investigation. These limitations define the priority agenda for future development and validation.

5.4 Future Research Directions

The most urgent research priority is implementation of clinical trials with actual T1D patients across diverse care settings. These are required to rigorously validate the effect of the system on glycaemic outcomes, patient satisfaction, and economic costs, and to refine the platform based on real patient feedback. Future physiological modelling work should extend coverage to a broader range of variables — including stress effects, acute and chronic illness, and more granular physical activity modelling — with particular priority on hormonal and menstrual cycle influences on insulin sensitivity, an area currently underrepresented in all major AID systems.

Advanced AI methods — specifically supervised and reinforcement learning modules — hold substantial potential for adaptive real-time insulin dosing and predictive analytics capable of anticipating and responding to individual patient requirements. Establishing these capabilities with transparency, explainability, and robust clinical oversight will be essential to building confidence among patients and clinicians. Integration with mobile platforms, CGM devices, insulin pumps, and electronic health records will further enhance accessibility and clinical workflow compatibility.

In summary, this thesis has demonstrated that a modular, data-driven platform can meaningfully improve Type 1 diabetes care — paving the way towards a personalised, accessible, and clinically integrated future. With a firm technical and clinical foundation established here, future research can narrow the gap between technological development and real-world patient experience, ultimately contributing to diabetes management that is effective, affordable, and patient-centred for all.

Clinical Reference: Halle M et al. Therapie des Typ-1-Diabetes, S3-Leitlinie, Version 5.0. AWMF-Registry 057-013. Diabetologie und Stoffwechsel 2024; 19: S155–S166. DOI: 10.1055/a-2312-0276.